Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

BASIC MATHEMATICS



DIBY (DO IT BY YOURSELF)-(01 to 60)

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DIBY-01

The multiplication of a rational number 'x' and an irrational number 'y' is

(A) Always rational **(B)** Rational except when $y = \pi$

(C) Always irrational **(D)** Irrational except when $x = 0$

[Ans. : (D)]

DIBY-02

If a, b, c are real, then $a(a-b)+b(b-c)+c(c-a)=0$, only if

(A) $a+b+c=0$ **(B)** $a=b=c$

(C) $a=b$ or $b=c$ or $c=a$ **(D)** $a-b-c=0$

[Ans. : (B)]

DIBY-03

If a, b, c are real and distinct numbers, then the value of $\frac{(a-b)^3+(b-c)^3+(c-a)^3}{(a-b)(b-c)(c-a)}$

(A) 1 **(B)** abc **(C)** 2 **(D)** 3

[Ans. : (D)]

DIBY-04

If $x^2 - 4x + 1 = 0$, then what is the value of $x^3 + \frac{1}{x^3}$?

[Ans. : 52]

DIBY-05

If $x = 1 + 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$, then $x^3 - 3x^2 - 3x =$ _____.

[Ans. : 1]

DIBY-06

If $x + \frac{1}{x} = a$, then what is the value of $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$?

[Ans. : $a^3 + a^2 - 3a - 2$]

DIBY-07

If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then $(x + y + z)^3$ is equal to

(A) 1 (B) 3 (C) $3xy$ (D) $27xyz$

[Ans. : (D)]

DIBY-08

The product $(x + y)(x - y)(x^2 + xy + y^2)(x^2 - xy + y^2)$ simplifies to

[Ans. : $x^6 - y^6$]

DIBY-09

Find the real values of p, q, r satisfying $(2p - 3)^8 + (1 - q)^6 + (4 - 3r)^4 = 0$.

[Ans. : $p = \frac{3}{2}, q = 1, r = \frac{4}{3}$]

DIBY-10

Let a, b and c be distinct non-zero real numbers such that $a+b+c=0$ then the value of the expression $\frac{(a+b)^2}{ab}+\frac{(b+c)^2}{bc}+\frac{(c+a)^2}{ca}$ is

(A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (D)]

DIBY-11

If $a+b+c=6$ & $a^2+b^2+c^2=14$ and $a^3+b^3+c^3=36$, then the value $3\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ of

[Ans. : 5.5]

DIBY-12

	Column-I		Column-II
(A)	$\frac{5x-3}{-4} \leq 8$	(P)	$x \in (-\infty, -3] \cup [2, 5)$
(B)	$\frac{x-1}{x+2} > 0$	(Q)	$x \in (-8. -4) \cup (-1, 3) \cup (9, \infty)$
(C)	$\frac{(x+3)(x-2)}{(x-5)} \leq 0$	(R)	$x \in \left[-\frac{29}{5}, \infty\right)$
(D)	$\frac{(x+4)(x+8)}{(x-9)(x+1)(x-3)} > 0$	(S)	$x \in (-\infty, -2) \cup (1, \infty)$

[Ans. : A → R, B → S, C → P, D → Q]

DIBY-13

(1) $\frac{6x-5}{4x+1} < 0$ **[Ans. :** $(-1/4, 5/6)$**]**

(2) $\frac{3}{x-2} < 1$ **[Ans. :** $(-\infty, 2) \cup (5, \infty)$**]**

(3) $\frac{1}{x-1} \leq 2$ **[Ans. :** $(-\infty, 1) \cup [3/2, \infty)$**]**

(4) $\frac{x-1}{x} - \frac{x+1}{x-1} < 2$ **[Ans. :** $(-\infty, -1)\left(0, \frac{1}{2}\right) \cup (1, \infty)$**]**

(5) $1 + \frac{2}{x-1} \geq \frac{6}{x}$ **[Ans. :** $(-\infty, 0) \cup (1, 2] \cup [3, \infty)$**]**

DIBY-14

Find sum of all integral values of x satisfying $\frac{x^2-5x+6}{x^2-x-12} \leq 0$

[Ans. : 03]

DIBY-15

Solve : $\frac{x}{x-2} > \frac{1}{x+3}$

[Ans. : $x \in (-\infty, -3) \cup (2, \infty)$]

DIBY-16

Solve : $\frac{2x+3}{x^2+x-12} \leq \frac{1}{2}$

[Ans. : $x \in (-\infty, -4) \cup [-3, 3) \cup [6, \infty)$]

DIBY-17

Solve : $(x+5)(2x-3)^5(-x+7)^3(3x+8)^2 < 0$

[Ans. : $x \in \left(-5, \frac{-8}{3}\right) \cup \left(\frac{-8}{3}, \frac{3}{2}\right) \cup (7, \infty)$**]**

DIBY-18

Solve : $(x^2-3x+2)(x^3-3x^2)(4-x^2) \geq 0$

[Ans. : $x \in (-\infty, -2) \cup \{0\} \cup [1, 3]$**]**

DIBY-19

Solve : $\frac{1}{x^2+x} \leq \frac{1}{2x^2+2x+3}$

[Ans. : $x \in (-1, 0)$**]**

DIBY-20

Solve : $\frac{(x-1)^2(x^2-4)(x^2-5x+6)}{(x^3+6x^2)} \geq 0$

[Ans. : $x \in (-6,-2] \cup \{1,2\} \cup [3,\infty)$**]**

DIBY-21

The greatest negative integer satisfying $x^2 - 4x - 77 < 0$ **and** $x^2 > 4$**, is**

[Ans. : -3**]**

DIBY-22

The solution set of $x^2 + 2 \leq 3x \leq 2x^2 - 5$**, is**

(A) **No Solution**

(B) $[1,2]$

(C) $(-\infty,-1] \cup [5/2,\infty)$

(D) **none of these.**

[Ans. : (A)]

DIBY-23

Solve: $-1 < \frac{3x-1}{x-2} \leq 4$

[Ans. : $x \in \left(-\infty, \frac{3}{4}\right) \cup [7, \infty)$]

DIBY-24

Values of x simultaneously satisfying: $x^2 - x - 6 \geq 0$ & $x^2 - 4x < 0$

[Ans. : $x \in [3, 4)$]

DIBY-25

Solve: $(x^2 - x - 1)(x^2 - x - 7) < -5$

[Ans. : $x \in (-2, -1) \cup (2, 3)$]

DIBY-26

Solution of $\frac{(x^4)(x+1)^2(x-2)}{(x-3)^3(x+4)} > 0$ **is**

(A) $x \in (-4, 2) \cup (4, \infty)$

(B) $x \in (-4, -1) \cup (-1, 2) \cup (3, \infty)$

(C) $x \in (-4, 0) \cup (0, 2) \cup (3, \infty)$

(D) $x \in (-4, -1) \cup (-1, 0) \cup (0, 2) \cup (3, \infty)$

[Ans. : (D)]

DIBY-27 [Challenger Question]

Solve: $\frac{(x^3-4x^2+3x-12)^{18}(5-7x)^{19}}{(4-x^2)^{20}(x+2)^{21}(x+3)^{22}} \leq 0$

[Ans. : $x \in (-\infty, -2) \cup \left[\frac{5}{7}, \infty\right) - \{-3, 2\}$**]**

DIBY-28

Solve following Inequalities :

(i) $\left(\frac{x^4+x^2+1}{x^2-11x+28}\right) \geq 0$

(ii) $\frac{x^2-x+1}{x^2+7x+15} \geq 0$

[Ans. : (i) $x \in (-\infty, 4) \cup (7, \infty)$**, (ii)** $x \in R$**]**

DIBY-29

If $a = 4^{\frac{-3}{2}}, b = (125)^{\frac{-2}{3}}, c = 81^{\frac{-3}{4}}$, then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} =$

[Ans. : 60]

DIBY-30

If $p = 8^{\frac{2}{3}}, q = 9^{\frac{3}{2}}, r = 16^{\frac{5}{4}}, s = \frac{1}{6^{-2}}$, then $p + q + r + s =$

[Ans. : 99]

DIBY-31

If $u = \left(\frac{1}{27}\right)^{\frac{-4}{3}}, v = \left(\frac{1}{216}\right)^{\frac{-2}{3}}, w = (8)^{\frac{5}{3}} \cdot (4)^{\frac{-5}{2}}, t = (9)^{\frac{3}{2}} \cdot (81)^{\frac{-3}{4}}$, then $\frac{vwt}{\sqrt{u}} =$

[Ans. : 4]

QUESTIONS ON LOG

DIBY-32

If $m_1 = \log_8 16, m_2 = \log_{81} 27, m_3 = \log_{1/3} 1/9, m_4 = \log_{1/3} 9\sqrt{3}$,

then $m_1 \cdot m_2 \cdot m_3 \cdot m_4$

[Ans. : –5]

DIBY-33

The value of $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$ is equal to

(A) 49 (B) 625 (C) 216 (D) 890

[Ans. : (D)]

DIBY-34

Let $N = \frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$. Then N is

(A) a natural number
(B) a prime number
(C) a rational number
(D) an integer

[Ans. : A, B, C, D]

DIBY-35

Greatest integer less than or equal to the number

$\log_2 15 \cdot \log_{1/6} 2 \cdot \log_3 1/6$ **is**

(A) 4 (B) 3 (C) 2 (D) 1

[Ans. : (C)]

DIBY-36

$\frac{1}{1+\log_b a+\log_b c} + \frac{1}{1+\log_c a+\log_c b} + \frac{1}{1+\log_a b+\log_a c}$ **has the value equal to**

(A) abc **(B)** $\frac{1}{abc}$ **(c) 0 (D) 1**

[Ans. : (D)]

DIBY-37

$\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc}$ **has the value equal to**

(A) 1/2 (B) 1 (C) 2 (D) 4

[Ans. : (B)]

DIBY-38

If $\log_7 2 = m$, then $\log_{49} 28$ is equal to

(A) $2(1+2m)$ (B) $\frac{1+2m}{2}$ (C) $\frac{2}{1+2m}$ (D) $1+m$

[Ans. : (B)]

DIBY-39

Value of $\log_2\left(\log_2\left(\log_3\left(\log_3 27^3\right)\right)\right) =$

[Ans. : 0]

DIBY-40

Match the column

	Column-I		Column-II
(A)	If $\log a = 2\log b$, then	(P)	$a^3 = 10b^2$
(B)	If $\log a + 2\log b = 0$, then	(Q)	$a = b$
(C)	If $3\log a - 2\log b = 1$, then	(R)	$a = b^2$
(D)	If $\ln\left(\frac{a+b}{2}\right) = \frac{1}{2}(\ln a + \ln b)$, then	(S)	$ab^2 = 1$

[Ans. : A–R, B–S, C–P, D–Q]

DIBY-41

If $x = \log_a(bc),\ y = \log_b(ca)$ and $z = \log_c(ab)$, then the value of $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1}$ = _____.

[Ans. : 1]

DIBY-42

If $N = 7^{\log_{49}900}, A = 2^{\log_2 4} + 3^{\log_2 4} + 4^{\log_2 2} - 4^{\log_2 3}$

$D = (\log_5 49)(\log_7 125)$. Find $P = \log_{\left(A - \frac{N}{10}\right)} |N + A + D + 6| - \log_5 2$.

[Ans. : 2]

DIBY-43

If $\log_{x-3}(2x-3)$ is a meaningful quantity then find the interval in which x must lie

(A) $x \in (3, 4] \cup (4, \infty)$

(B) $x \in [3, 4) \cup (4, \infty)$

(C) $x \in (3, 4) \cup (4, \infty)$

(D) None of these

[Ans. : (C)]

DIBY-44

Suppose $x, y, z > 0$ and different than one and $\ln x + \ln y + \ln z = 0$.

If $\frac{1}{e^k} = x^{\frac{1}{\ln y}+\frac{1}{\ln z}} \cdot y^{\frac{1}{\ln z}+\frac{1}{\ln x}} \cdot z^{\frac{1}{\ln x}+\frac{1}{\ln y}}$. The k =

[Ans. : 3]

DIBY-45

Let $\log_2 x + \log_4 x + \log_8 x = \log_k x$ for all $x \in R^+$. If $k = \sqrt[b]{a}$ where $a, b \in N$, then find the smallest positive value of $(a + b)$.

[Ans. : 75]

DIBY-46

If $A = \log_{\sqrt{5}} \left(\left(5^{\frac{1}{2}}\right)^{\frac{1}{2}}{}^{\cdot^{\cdot^{\cdot^{\frac{1}{2}}}}}\right)$ .10 times , then value of $\log_{\sqrt{3}}(1024A + 1)$, is equal to

(A) 1 (B) 3 (C) 5 (D) 2

[Ans. : (D)]

DIBY-47

Value of x satisfying: $x^2 + 7^{\log_7 x} - 2 = 0$

Ans: (x = 1)

DIBY-48

The sum of all the solutions of the equation: $2\log x - \log(2x - 75) = 2$

(A) 30 (B) 350 (C) 75 (D) 200

Ans: D

DIBY-49

The solution of the equation: $\log_7 \log_5 \left(\sqrt{x^2 + 5 + x}\right) = 0.$

(A) x = 2 (B) x = 3 (C) x = 4 (D) x = −2 Ans: C

DIBY-50

Number of real solution of: $\log_5[2+\log_3(x+3)]=0$ is

Ans: 1

DIBY-51

The sum of all integral values of x satisfying the equation

$2\log_8(2x)+\log_8(x^2-2x+1)=\frac{4}{3}$ is.

[Ans. : 2]

DIBY-52

$\log_7\left(\frac{2x-6}{2x-1}\right)\geq 0$

Ans: $\left(-\infty,\frac{1}{2}\right)$

DIBY-53

If $\log_{0.5}\log_5(x^2-4) > \log_{0.5}1$, then '$x$' lies in the interval

(A) $(-3,-\sqrt{5}) \cup (\sqrt{5},3)$ **(B)** $(-3,-\sqrt{5}) \cup (\sqrt{5},3\sqrt{5})$

(C) $(\sqrt{5},3\sqrt{5})$ **(D)** ϕ

[Ans. : (A)]

DIBY-54

Solution set of the inequality $2-\log_2(x^2+3x) \geq 0$ is

(A) $[-4,1]$ **(B)** $[-4,-3] \cup (0,1]$

(C) $(-\infty,-3) \cup (1,\infty)$ **(D)** $(-\infty,-4) \cup [1,\infty)$

[Ans. : (B)]

DIBY-55

Number of solution of the equation $\log_3(3^x-8) = 2-x$ is

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

[Ans. : (B)]

DIBY-56

Number of solution of the equation $\log_{x-1}(4) = 1 + \log_2(x-1)$ is

(A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (C)]

DIBY-57

Number of solution of the equation $(x+1)^{\log_{10}(x+1)} = 100(x+1)$ is

(A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (C)]

DIBY-58

$$\log_{\frac{3}{4}}\log_8\left(x^2+7\right) + \log_{\frac{1}{2}}\log_{\frac{1}{4}}\left(x^2+7\right)^{-1} = -2$$

[Ans. : x = 3 or –3]

DIBY-59

The product of all positive real values of x satisfying the equation

$x^{\left(16(\log_5 x)^3 - 68\log_5 x\right)} = 5^{-16}$ is

[JEE (Adv.)-2022]

[Ans. : 1]

DIBY-60

Let (x_0, y_0) be the solution of the following equations

$(2x)^{\ln 2} = (3y)^{\ln 3}$

$3^{\ln x} = 2^{\ln y}$

Then x_0 is [JEE (Adv.)-2011]

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 6

[Ans. : (C)]

It is never about end result

It's always about Journey

THANK You

#futureIITians